## Selection Sort

Barometer statement: $a[j] < a[min]$

$$\sum_{i=1}^{n-1} \sum_{j=i+1}^{n} 1 = n^2 - \frac{n(n+1)}{2} \in O(n^2)$$

## Insertion Sort

Bar. stm.: $j = j - 1$

$$\sum_{i=2}^{n} \sum_{j=1}^{i-1} 1 = \frac{n^2}{2} - \frac{n}{2}$$

## Bubble Sort

B.S.: $a[j-1] > a[j]$

$$\sum_{i=1}^{n} \sum_{j=i}^{n} 1 = \sum_{i=1}^{n} (i-1) = \sum_{i=1}^{n} i - \sum_{i=1}^{n} 1$$

$$\frac{n \cdot (n+1)}{2} - n = \frac{n^2}{2} - \frac{n}{2}$$

## Heap Sort

for loop n times
  delete min $O(\log n)$

$O(n \cdot \log n)$

## Merge Sort

divide and conquer

## Quick Sort

## Hashing

| (arrays) | insert | remove | search |
|---|---|---|---|
| Ordered | n | n | log n |
| Unordered | 1 | n | n |

$f(key) = key$ direct addressing

data size $\gg$ table size

Cost of unsuccess search $\frac{1}{1-a}$ unsuccess search $\frac{m}{m-i} = \frac{1}{1 - \frac{i}{m}}$ arr. index

# of probes for succ. search $t(n) = \frac{1}{a} \cdot \ln \frac{1}{1-a}$

Quadratic probing $h'(a_k, i) = (h(a_k) + c_1 i + c_2 i^2) \bmod m$

Popular form of $h_2$ $h_2(a_k) = R - (a_k \bmod R)$

## Horner's Rule

$$\sum_{i=0}^{keysize-1} key[keysize-i-1] \times 37^i \bmod tablesize$$

Multiplication method: $h(k) = \lfloor m \cdot (k \cdot A \bmod 1) \rfloor$
$0 < A < 1$

Seperate chaining: last digit of k → ✱ $k = a[x]$

Double hashing $h'(a_k, i) = (h_1(a_k) + i\, h_2(a_k)) \bmod m$

Rehashing: if load 50+% new and larger $m_{new}$ = the first prime $> 2 m_{old}$

Folding method: 5103 + 24 = 78 → 1072 → 5 4 → 60

k = 510324

Midsquare method: $k = 510324$
$k^2 = 260430\,58\,1976$

load factor $(a) = \frac{n}{m} = \frac{\text{occupied cells}}{\text{table size}}$

Linear probing $h'(a_k, i) = (h(a_k) + i) \bmod m$ $i$ = # of coll.

R = the closest prime smaller to tablesize m

$C_{ins} = \frac{1}{1-a}$ $C_{search} = \frac{1}{a} \cdot \ln \frac{1}{1-a}$

## GRAPHS

Adjecency Matrix R. space req. $O(|V|^2)$ − waste of space for sparse graphs + fast access

Adjecency List R. space req. $O(|E|+|V|)$ − sequential search + minimum space

i → source vertex: 1st destination 2nd weight

Topological sort: is an ordering of vertices Kruskal's A. $E \log E$

Breadth First Search time $E+V$ · Depth first search time: $E+V$

Dijkstra's Algorithm $E \log V$ min-heap Prim's A. $E \log V$

Prime Numbers up to 100: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41
43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

English Alphabet: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T
U, V, W, X, Y, Z

## Time Complexity

| | Average Access | Search | Insertion | Delete | Worst Access | Search | Insert | Delete | Space Complexity |
|---|---|---|---|---|---|---|---|---|---|
| Array | 1 | 3 | 3 | 3 | 1 | 3 | 3 | 3 | 3 |
| Stack | 3 | 3 | 1 | 1 | 3 | 3 | 1 | 1 | 3 |
| Queue | 3 | 3 | 1 | 1 | 3 | 3 | 1 | 1 | 3 |
| Singly LinkedList | 3 | 3 | 1 | 1 | 3 | 3 | 1 | 1 | 3 |
| Doubly Linked List | 3 | 3 | 1 | 1 | 3 | 3 | 1 | 1 | 3 |
| Hash Table | | 1 | 1 | 1 | | 3 | 3 | 3 | 3 |
| BST | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 3 | 3 |
| B-Tree | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 |
| Splay Tree | | 2 | 2 | 2 | | 2 | 2 | 2 | 3 |
| AVL Tree | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 |

$1 \Rightarrow 1$ $\quad 3 \Rightarrow n$ $\quad 5 \Rightarrow n^2$ $\quad 7 \Rightarrow n!$
$2 \Rightarrow \log n$ $\quad 4 \Rightarrow n \cdot \log n$ $\quad 6 \Rightarrow 2^n$

Merge two sorted list $n, m$ nodes $n+m-1$ comparison

Avg. LL search $\frac{l+1}{2}$

zig = L
zag = R
RL-LR double inside

AVL: $|h_{N_L} - h_{N_R}| \leq 1$
RL right subt. of left child

## Time Complexity

| | Best | Average | Worst | Space Complexity |
|---|---|---|---|---|
| Quicksort | 4 | 4 | 5 | 2 |
| Mergesort | 4 | 4 | 4 | 3 |
| Heapsort | 4 | 4 | 4 | 1 |
| Bubblesort | 3 | 5 | 5 | 1 |
| Insertionsort | 3 | 5 | 5 | 1 |
| Selectionsort | 5 | 5 | 5 | 1 |
| Treesort | 4 | 4 | 5 | 3 |
| Countingsort | | | | |

AVL min of #node: $S(h) = S(h-1) + S(h-2) + 1$
$S(0) = 1 \quad S(1) = 2$
Best case BST: $\lfloor \log_2 n \rfloor + 1$ $\quad f(n) = 2f(n/2) + n - 1$

## B-Trees

$keys_{min}: t-1$ $\quad keys_{max}: 2t-1$

every internal node can have $2t$ children max.

For a B-tree min degree $t \geq 2$ and $n$ nodes

$$h \leq \log_t \frac{n+1}{2}$$

## Heaps

If complete BT in array element in $i^{th}$ then right c.: $2i+1$ left c.: $2i$

Binary heap insertion $O(\log n)$ (worst/average/best) delete min $O(\log n)$ search $O(n)$ find min $O(1)$

build heap: avg $n$ worst $n \cdot \log n$

## Leftist Heaps

not p. balanced
merging $O(\log n)$
binary tree

NPL(NULL) = -1
NPL(LEAF) = 0

NPL: length shortest path $x \to$ without 2 child
NPL(LC(X)) $\geq$ NPL(RC(X))
insert/remove/delete min: $O(\log n)$

Suppose two leftist heaps $L_1, L_2$ and $n$ elements total how many comp. to merge $L_1, L_2$?

a) Worst case
$L_1 = n/2$ nodes $\quad n/2 = 2^{k-1} - 1$
k-1 nodes at right most path $\quad 2(k-1) - 1 = 2k - 3$
$k = \log_2(n+2)$ $\quad N = 2k - 3$

b) Best case $N = 1$

$2^4 = 16$ $\quad 2^7 = 128$ $\quad 2^{10} = 1024$ $\quad 2^{13} = 8192$ $\quad 2^{16} = 65536$
$2^5 = 32$ $\quad 2^8 = 256$ $\quad 2^{11} = 2048$ $\quad 2^{14} = 16384$ $\quad 2^{17} = 131072$
$2^6 = 64$ $\quad 2^9 = 512$ $\quad 2^{12} = 4096$ $\quad 2^{15} = 32768$ $\quad 2^{18} = 262144$

## Binomial Heaps

$O(1)$ for insert
$O(\log n)$ for others
$B_0 \to$ single node
$B_{k-1} + B_{k-1} \to B_k$ Binomial tree $B_k$
height $= k$ $\quad n = 2^k$ nodes $\quad k+1$ depth levels $[0, k]$
Ordered $B_0 B_1 B_2 \cdots B_{k-1}$ $\quad C(k, d]$ nodes depth level $d$

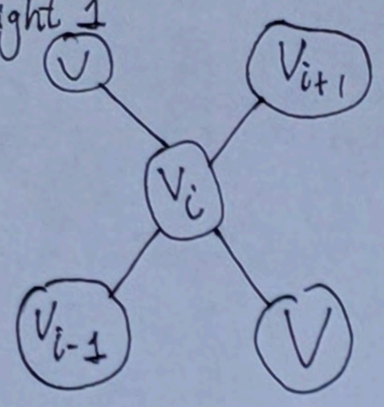
$$\sum_{k=1}^{n} r^{k-1} = 1 + r + r^2 + r^3 + \cdots + r^{n-1} = \frac{1 - r^n}{1 - r} \quad (r \neq 1)$$

$n = 2^k \to k = \log_2(n+1)$

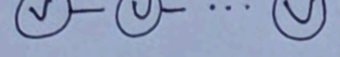
most crowded $\lceil k/2 \rceil$ depth level of $B_k$

All edges have weight 1

MST minimize weight sum



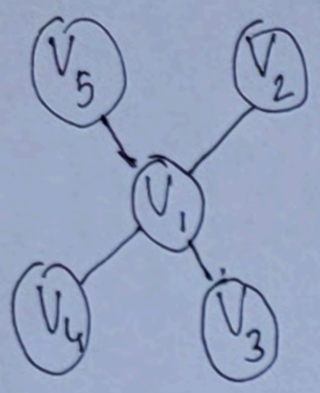

MST maximize weight sum




weight sum $= s(v_i, v_j)$ sum of weights of edges forming path between $v_i$ and $v_j$

weight sum between all pairs $S$

$$S = \sum_{i=1}^{n} \sum_{j=1 j\neq i}^{n} s(v_i, v_j)$$

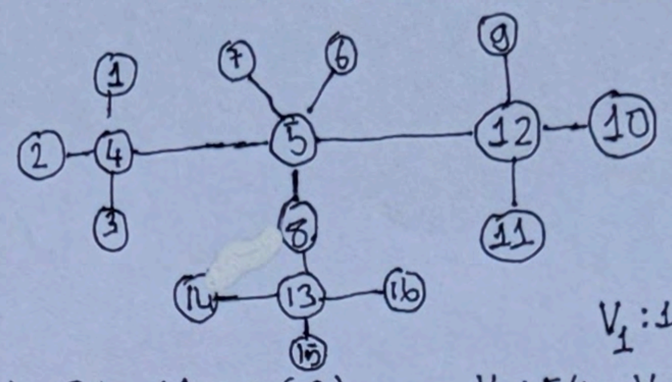



| | | | | |
|---|---|---|---|---|
| 1,2:1 | 2,1:1 | 2,3:2 | 3,4:2 | 4,5:2 |
| 1,3:1 | 3,1:1 | 2,4:2 | 3,5:2 | 5,2:2 |
| 1,4:1 | 4,1:1 | 2,5:2 | 4,2:2 | 5,3:2 |
| 1,5:1 | 5,1:1 | 3,2:2 | 4,3:2 | 5,4:2 |

total weight sum $2(n-1)$

$2(n-1)(n-2)$

total weight $(S1 = 2(n-1)(n-2) + 2(n-1) \in O(n^2)$




uniformly weighted edges

one of MST

minizes avg. distance between all pair of vertices

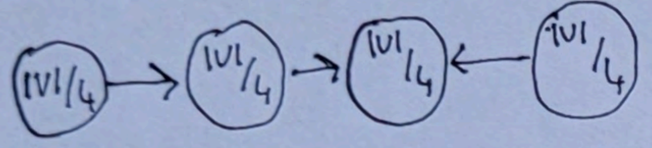
$V_1: 1/24$ $V_2: 2/23$ $V_3: 3/22$ $V_4: 4/21$

$V_5: 5/20$ $V_6: 6/19$ $V_7: 7/18$ $V_8: 8/17$

$V_{13}: 9/16$ $V_{14}: 10/15$ $V_{15}: 11/14$ $V_{16}: 12/13$

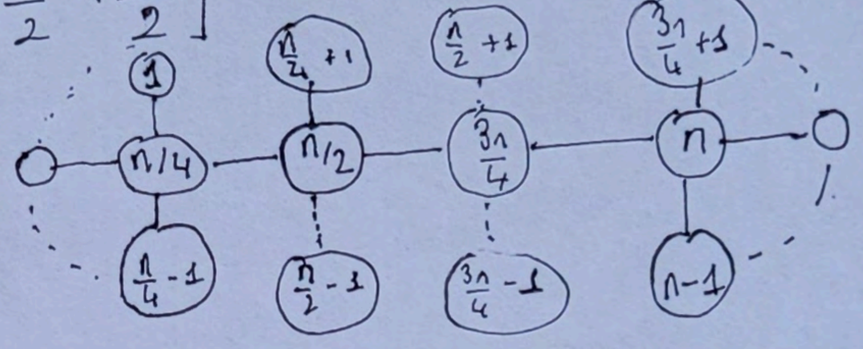
$V_9: 25/32$ $V_{10}: 26/31$ $V_{11}: 27/30$ $V_{12}: 28/29$

$V_1 - V_2 - V_3 \cdots V_4$ $\quad S = 2\left[\frac{n(n-1)}{2} + \frac{(n-1)(n-2)}{2} + \frac{(n-2)(n+3)}{2} + \cdots + \frac{3\cdot 2}{2} + \frac{2\cdot 1}{2}\right]$




each subgraph has quarter of all vertices

Weakly connected. For instance vertices at the leftmost subg. aren't acces. from any other.




$|V| = n$

MST

Two binomial queues $BQ_1$ $BQ_2$

$n = 2^k$ $k \leq e$ elements $m = 2^e$

$BQ_1: \{B_k\}$ $BQ_2: \{B_L\}$ —delete min→ $BQ_1: \{B_0 B_1 \cdots B_{k-1}\}$ $BQ_2: \{B_0, B_1 \cdots B_{e-1}\}$ —merge→ $BQ_M: \{B_1 B_2 B_3 \cdots B_{k1} B_k B_{k+1} \cdots B_e\}$

$BQ_1: \{B_k\}$ —merge→ $BQ_M: \{B_k B_e\}$ → $BQ_M: \{B_0 B_1 B_2 \cdots B_{k-1} B_e\}$

$BQ_2: \{B_e\}$ delete min → 2nd delete min → $BQ_M: \{B_0 B_1 B_2 \cdots B_{k-2} \{B_0 B_1 \cdots B_{k-2}\} B_e\}$ ($B_k$ after delete min)

both min keys is $B_{1L}$

the second root of most crowded $B_{k-1}$

merge same heights

$BQ_M: \{B_1 B_2 \cdots B_{k-2} B_{k-1} B_e\}$

ascending: max-heap

descending: min-heap

$B_1$ $B_0$ $B_1$ $B_2$ $B_3$ $B_4$

$B_0$ $B_4$ $B_5$ $B_6$

$B_2$ $B_3$ $B_4$ ⋯

$B_4$ $B_5$ $B_6$ ⋯

$B_0$ $B_0$ $B_1$ $B_2$ $B_3$ $B_4$ $B_4$ $B_5$ $B_5$ $B_6$ $B_6$ $B_7$ $B_8$ $B_9$

$B_1$ $B_2$ $B_3$

$B_4$ $B_4$ $B_4$ $B_5$ $B_5$ $B_6$ $B_6$ $B_7$ $B_8$ $B_9$

$B_6$ $B_7$ $B_7$ $B_8$ $B_9$ $B_{10}$

$\frac{n+1-4}{1}$

$4 \cdot \frac{(n+1)(n+2)}{2}$

$\binom{5}{3}$ $5!$

$\frac{5 \cdot 4 \cdot 3}{3 \cdot 2} = 10$

$\frac{12!}{(5-3)! \cdot 3!}$

$36 = 3^2 \cdot 2^2$

$\frac{4!}{2 \cdot 2!}$ $\frac{24}{7} = 2$

$1 - \frac{1}{4}$

$\frac{4 \cdot 3}{2}$

$n$ $2n$ $\frac{n-1}{2}$

$n-1$ $\frac{3n+1}{2}$

$n-2$ $n+1$

$n + \frac{n-1}{2}$

$\frac{3n-1}{2}$

$n'$

$G_1$ $G_2$ $G_3$ $G_4$ $G_5$ $G_6$

$\frac{3n-1}{2}$

$n$

$\frac{n+1}{2}$

$6n + \frac{n-1}{2} = \frac{13n-1}{2}$

$6n + \frac{n-1}{2} = \frac{13n+1}{2}$

$(4n+1) + \frac{n-1}{2}$

$\frac{3n+1+n-1}{2}$ $\quad 4n+2+n-1$ $\quad 5n+1$ $\quad \frac{9n+1}{2}$

$\frac{3n+1+n-1}{2}$ $\quad 6n+2+n-1$ $\quad 7n+1$ $\quad 5n+1+\frac{n-1}{2}$

$10n+2+n-1$